

PART - A (ii) (3 × 3 = 9 Marks)

- 7 Illustrate how the basic principles of FOSS development can be used to enable collaborative software development.
- 8 Explain how a boot loader functions during the Linux booting process
- 9 Analyze the effect of push and shift operations in the given list:
`my @data = (5, 10);
 push(@data, 15);
 shift(@data);
 print "@data";`

K-	CO
Level	
K3	CO1
K3	CO2
K4	CO3

PART - B (3 × 10 = 30 Marks)

10. a Compare Free Software and Open Source Software by applying their principles to real software projects.
- (Or)
- b Analyse a real-world software project and identify how a FOSS licensing model is implemented.
11. a Illustrate the steps involved in setting up a dual-boot system with Linux and another OS.

Mark	K-	CO
Split up	Level	
10	K3	CO1

- (Or)
- b Demonstrate Linux process management using system calls such as fork(), exec(), and wait()
- 12 a Write a Perl program that performs the following tasks:
1. Declare an array of 12 integers representing monthly sales figures.
 2. Use a loop to iterate through the array.
 3. For each month, classify the sales figure using if-elsif-else as:
 - High (greater than 1000)
 - Medium (between 500 and 1000, inclusive)
 - Low (less than 500)
 4. Print each month's sales along with its classification.
 5. At the end, display the total number of months in each category.

10	K3	CO1
10	K3	CO2
10	K3	CO2

- (Or)
- b Develop a Perl script using scalars, arrays, and hashes for a real-time application.

10	K3	CO2
----	----	-----

PART – C (1 × 15 = 15 Marks)

13. i) Analyse and implement a perl program to read a series of number and determine the frequency of even and odd numbers using looping and decision making statements.

Mark Split up	K- Level	CO
8	K4	CO3

7

- ii) Write a perl program to generate multiplication table for a given number using loops and apply conditional logic to highlight specific multiples. (For eg: highlight multiples of 5).

Course Outcomes (with Knowledge Level)

CO1	Apply the fundamental concepts of Free and Open Source Software and licensing models to select appropriate open-source software.	K3
CO2	Demonstrate the use of Linux operating system features for installation, configuration, and system administration tasks.	K3
CO3	Examine Perl programs by analysing program structure, data types, and control statements.	K4
CO4	Develop Perl programs using suitable programming constructs to solve practical problems.	K3
CO5	Evaluate open-source development tools such as version control, bug tracking, and package management systems.	K4
CO6	Interpret Free and Open Source Software case studies to understand open-source project development practices.	K4

Distribution of COs (Percentage wise)

CO No.	CO1	CO2	CO3	CO4	CO5	CO6
%	50	25	25	-	-	-



SAI RAM INSTITUTE OF TECHNOLOGY

An Autonomous Institution | Affiliated to Anna University & Approved by AICTE, New Delhi
Accredited by NBA and NAAC 'A+' | BIS/EOMS ISO 21001 : 2018 Certified and NIRF ranked institution
Sai Leo Nagar, West Tambaram, Chennai - 600 044. www.sairamit.edu.in



CONTINUOUS ASSESSMENT TEST - I

Question Paper Code

26AU81F093

Course Code - Name : 20ITEL801 - Free and Open Source Software Tools

Degree & Program : B.Tech & IT

Level

: Intermediate

Date of Exam : 10.02.2026

Year / Semester

: IV/VIII

Duration : 1 Hour 45 Minutes

Max. Marks

: 60

Answer ALL Questions

PART - A (i) ($6 \times 1 = 6$ Marks)

- | | | K-
Level | CO |
|---|--|-------------|-----|
| 1 | In which phase do the users who require the software to fall into?
a) Download and Install b) End-Use
c) Communicate Experience d) None of these | K1 | CO1 |
| 2 | Which firewall is most commonly used on Linux?
a) ipchains b) ipfw c) pf d) iptables | K1 | CO2 |
| 3 | What scope does the 'our' keyword provide when declaring variables in Perl?
a) Package-wide scope b) Local scope
c) Temporary scope d) No specific scope | K1 | CO3 |
| 4 | Identify the minimum memory required to run a Linux system and explain why this amount is necessary. a) 2 Megabyte (Mb) b) 4 Megabyte (Mb)
c) 16 Megabyte (Mb) d) 64 Megabyte (Mb) | K2 | CO1 |
| 5 | LILLO is used during the system startup process. What best describes its function?
a) Listing all boot loaders b) Loading the operating system at boot time
c) Listing all hardware devices d) Displaying memory locations | K2 | CO2 |
| 6 | Which operator is used to specify a range of values in programming?
a) \$ b) .. c) \$_ d) \$1 | K2 | CO3 |

PART - A (ii) ($3 \times 3 = 9$ Marks)

- | | | K-
Level | CO |
|---|--|-------------|-----|
| 7 | Explain how community-based software development benefits projects by giving examples. | K3 | CO1 |
| 8 | Illustrate the process of configuring the X Window System, highlighting key components and settings. | K3 | CO2 |
| 9 | Differentiate between Simple Lists and Complex Lists by analyzing their composition, indexing, and typical applications. | K4 | CO3 |

PART – B (3 × 10 = 30 Marks)

		Mark Split up	K- Level	CO
10.	a	10	K3	CO1
	Illustrate different FOSS licensing models and demonstrate how the copy left principle is applied in real-world open-source projects.			
	(Or)			
	b	10	K3	CO1
	Compare GPL, AGPL, and LGPL licenses and analyze their applicability using suitable real-world examples.			
11.	a	10	K3	CO2
	Explain how processes are created, managed, and ended in Linux. Show examples of common process management system calls.			
	(Or)			
	b	10	K3	CO2
	Explain the X Window System architecture and show how to configure it step by step in Linux.			
12.	a	10	K3	CO3
	Apply decision-making and looping statements to solve a programming problem, using appropriate examples.			
	(Or)			
	b	10	K3	CO3
	Demonstrate the use of lists and hashes by performing common operations with suitable examples.			

PART – C (1 × 15 = 15 Marks)

		Mark Split up	K- Level	CO
13.	(i) Create a perl program to generate a right angled triangle number pattern using loops and apply decision statements to replace all prime number with * symbol in the pattern. Number pattern is given below for reference:	8	K4	CO3
	<pre> 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5 </pre>			
	(ii) Write a perl program to process a list of strings using loops and apply decision logic to separately count uppercase letters, lower case letters and digits.	7		



SAI RAM INSTITUTE OF TECHNOLOGY

An Autonomous Institution | Affiliated to Anna University & Approved by AICTE, New Delhi
Accredited by NBA and NAAC 'A+' | BIS/EOMS ISO 21001 : 2018 Certified and NRF ranked Institution
Sai Leo Nagar, West Tambaram, Chennai - 600 044, www.sairamit.edu.in



CONTINUOUS ASSESSMENT TEST - II

Question Paper Code	26AU82-092
---------------------	------------

Course Code - Name : 20ITEL801 - Free And Open-Source Software Tools
Degree & Program : B. Tech IT Level : Intermediate
Date of Exam : 16. 03. 2026 Year / Semester : IV/VIII
Duration : 1 Hours 45 Minutes Max. Marks : 60

Answer ALL Questions

PART - A (i) (6× 1 = 6 Marks)

- | | K-Level | CO |
|---|---------|-----|
| 1 Which statement is used to enable strict mode in Perl?
a) Strict mode; b) use strict; c) use strict mode; d) None of these | K1 | CO4 |
| 2 Name the tool used mainly for bug tracking in software development?
a)Git b)Bugzilla c)Libreoffice d)GCC | K1 | CO5 |
| 3 Argo UML primarily supports which type of software development method?
a) Agile b) Waterfall c) Object-Oriented Design d) Spiral | K1 | CO6 |
| 4 Which keyword is used to define a subroutine in perl?
a)function b)method c)sub d)define | K2 | CO4 |
| 5 Which system is used to manage and install software packages?
a)compiler b)package manager c)debugger d)editor | K2 | CO5 |
| 6 Samba is used to implement which protocol?
a) FTP b) SMB (Server Message Block) c) HTTP d) IMAP | K2 | CO6 |

PART - A (ii) (3× 3 = 9 Marks)

- | | K-Level | CO |
|--|---------|-----|
| 7 Demonstrate how the fetchrow_array() API is used to retrieve data from a database query result in a program. | K3 | CO4 |
| 8 Demonstrate the four major operations used in a distributed version control system using Git. | K3 | CO5 |
| 9 Analyze the authentication and authorization mechanisms used by Samba to manage secure access to shared files and printers in a network environment. | K4 | CO6 |

PART - B (3 × 10 = 30 Marks)

- | | | Mark
Split
up | K-
Level | CO |
|-------|---|---------------------|-------------|-----|
| 10. a | Apply the concept of Common Gateway Interface to develop a simple web-based application and explain its working with an example. | 10 | K3 | CO4 |
| | (Or) | | | |
| b | Apply various error handling techniques in Perl to manage runtime errors with suitable examples. | 10 | K3 | CO4 |
| 11. a | Draw activity diagram, package diagram, class diagram for Hospital Management System. | 10 | K3 | CO5 |
| | (Or) | | | |
| b | Illustrate the role of bug tracking system and package managers in open software development. Compare at least two widely used bug tracking system. | 10 | K3 | CO5 |
| 12. a | Illustrate various standards and policies of open-source software development with suitable examples from LibreOffice and samba. | 10 | K3 | CO6 |
| | (Or) | | | |
| b | Demonstrate various features of Samba with examples | 10 | K3 | CO6 |

PART - C (1 × 15 = 15 Marks)

- | | | Mark
Split
up | K-
Level | CO |
|--------|--|---------------------|-------------|-----|
| 13. i) | Analyze the structure, modules and collaborative development practices of Libre Office and discuss how these aspects support efficient open source software development. | 15 | K4 | CO6 |

Course Outcomes (with Knowledge Level)

CO1	Apply the fundamental concepts of Free and Open Source Software and licensing models to select appropriate open-source software.	K3
CO2	Demonstrate the use of Linux operating system features for installation, configuration, and system administration tasks.	K3
CO3	Examine Perl programs by analyzing program structure, data types, and control statements.	K4
CO4	Develop Perl programs using suitable programming constructs to solve practical problems.	K3
CO5	Evaluate open-source development tools such as version control, bug tracking, and package management systems.	K4
CO6	Interpret Free and Open Source Software case studies to understand open-source project development practices.	K4

Distribution of COs (Percentage wise)

CO No.	CO1	CO2	CO3	CO4	CO5	CO6
%	-	-	-	44	27	27



SAI RAM INSTITUTE OF TECHNOLOGY

An Autonomous Institution | Affiliated to Anna University & Approved by AICTE, New Delhi
Accredited by NBA and NAAC 'A+' | BIS/EOMS ISO 21001 : 2018 Certified and NIRF ranked institution
Sai Leo Nagar, West Tambaram, Chennai - 600 044. www.sairamit.edu.in



CONTINUOUS ASSESSMENT TEST – II

Question Paper Code

26AU82A093

Course Code - Name : 20ITEL801 - Free and Open-Source Software Tools

Degree & Program : B. Tech IT

Level : Intermediate

Date of Exam : 23/03/2026(AN)

Year / Semester : IV/VIII

Duration : 1 Hour 45 Minutes

Max. Marks : 60

Answer ALL Questions

PART – A (i) (6× 1 = 6 Marks)

- 1 What is the correct operator precedence for the following operators?
&&, &, =, ->
a) &, &&, =, -> b) ->, &, &&, = c) =, &, &&, -> d) =, ->, &&, &
- 2 Which of the following is a version control system?
a) Eclipse b) Git c) MySql d) Apache
- 3 Samba provides interoperability between which of the following?
a) Linux and macOS b) Linux/Unix and Windows
c) Linux and Android d) macOS and Windows Answer:
- 4 Which keyword is used to define a subroutine in perl?
a)function b)method c)sub d)define
- 5 Which system is used to manage and install software packages?
a)compiler b)package manager c)debugger d)editor
- 6 What type of server can Samba enable Linux/Unix systems to act as?
a) Web server b) File server c) DNS server d) Email server

K-
Level
K1 CO

K1 CO

K1 CO

K2 CO4

K2 CO5

K2 CO6

PART – A (ii) (3× 3 = 9 Marks)

K-
Level
CO

- 7 Demonstrate how the `fetchrow_array()` API is used to retrieve data from a database query result in a program.

K3 CO4

- 8 Illustrate the suitability of open-source software for various projects and organizational environments with relevant examples.

K3 CO5

- 9 Demonstrate how Samba is used to enable file and printer sharing between Microsoft Windows and Linux systems.

K4 CO6

PART – B (3 × 10 = 30 Marks)

		Mark Split up	K- Lev el	CO
10.	a	10	K3	CO4
	i. Write a perl subroutine reverse_string that takes string as input and returns the reversed string. Demonstrate the subroutine with the string "OpenSource".			
	ii. Develop a perl subroutine remove_duplicates that removes duplicate elements from an array and returns unique elements. Demonstrate with the array (1,2,2,3,4,4,5)			
	(Or)			
	b	10	K3	CO4
	Demonstrate how a perl program can be used to interact with a MySQL database named library_db through the perl DBI module. Create a table books(book_id,title,author,price) if it does not exist, insert at least three sample records into table, retrieve and display the details of all books whose price is greater than 500.			
11.	a	10	K3	CO5
	Draw activity diagram ,package diagram, class diagram for Hospital Management System.			
	(Or)			
	b	10	K3	CO5
	Demonstrate how version control is used in Free and Open-Source Software development using tools like Git..			
12	a	10	K3	CO6
	Illustrate various standards and policies of open-source software with example.			
	(Or)			
	b	10	K3	CO6
	Demonstrate the key features of Samba used for file and printer sharing between Microsoft Windows and Linux systems with examples.			

PART – C (1 × 15 = 15 Marks)

		Mark Split up	K- Level	CO
13.	i	15	K4	CO6
	Analyze the architecture of Samba by examining its key components, benefits, and control features in managing network resources.			

Course Outcomes (with Knowledge Level)

CO1	Apply the fundamental concepts of Free and Open Source Software and licensing models to select appropriate open-source software.	K3
CO2	Demonstrate the use of Linux operating system features for installation, configuration, and system administration tasks.	K3
CO3	Examine Perl programs by analyzing program structure, data types, and control statements.	K4
CO4	Develop Perl programs using suitable programming constructs to solve practical problems.	K3
CO5	Evaluate open-source development tools such as version control, bug tracking, and package management systems.	K4
CO6	Interpret Free and Open Source Software case studies to understand open-source project development practices.	K4

Distribution of COs (Percentage wise)

CO No.	CO1	CO2	CO3	CO4	CO5	CO6
%	-	-	-	44	27	27